

Sabee Grewal

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CURRENT POSITIONS	Herman Goldstine Memorial Postdoctoral Fellow IBM Research	2026 – Present
	Postdoctoral Research Scientist Columbia University Host: Henry Yuen	2026 – Present
RESEARCH INTERESTS	quantum algorithms; quantum learning theory; quantum complexity theory; theoretical computer science	
EDUCATION	The University of Texas at Austin <i>PhD</i> in Computer Science Thesis: Efficient Learning of Quantum States and Circuits Advisor: Scott Aaronson	2021 – 2026
	The University of Texas at Austin <i>Master of Science</i> in Electrical and Computer Engineering	2021
	The University of Texas at Austin <i>Bachelor of Science</i> in Physics	2019
	The University of Texas at Austin <i>Bachelor of Science</i> in Electrical and Computer Engineering	2017
RECOGNITION	IBM PhD Fellowship	2024
	ITCS '23 Best Student Paper For “Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom”	2023
	Teaching Excellence Award Awarded by the Cockrell School of Engineering	2020
	Teaching Excellence Award Awarded by the Cockrell School of Engineering	2015
	Teaching Excellence Award Awarded by the Cockrell School of Engineering	2014
PUBLICATIONS	Sabee Grewal and Dorian Rudolph. “ QMA Has Perfect Completeness ”. <i>Preprint</i> .	
	John Bostanci, Sabee Grewal, Jonas Haferkamp, Andrew Huang, Yeongwoo Hwang, Anand Natarajan, and Chinmay Nirkhe. “ A Quantum Oracle Separation Between QMA(2) and QMA ”. <i>Preprint</i> .	
	Omar Alrabiah, Srinivasan Arunachalam, Sabee Grewal, and John Wright. “ No Low-Degree Tests for Quantum States ”. <i>Preprint</i> .	
	Sabee Grewal, Meghal Gupta, William He, Aniruddha Sen, and Mihir Singhal. “ Nearly Time-Optimal Pure State Tomography with Pauli Measurements ”. In: <i>FOCS 2026</i> .	
	Sabee Grewal and Dorian Rudolph. “ On the Pure Quantum Polynomial Hierarchy and Quantified Hamiltonian Complexity ”. In: <i>ICALP 2026</i> . Presented at TQC 2026.	

Sabee Grewal and Daniel Liang. “Efficient Learning of Structured Quantum Circuits via Pauli Dimensionality and Sparsity”. *Preprint*.

Sabee Grewal and William Kretschmer. “Unentanglement and Post-Measurement Branching in Quantum Interactive Proofs”. In: *TQC 2026*.

William Kretschmer, Sabee Grewal, Matthew DeCross, Justin A. Gerber, Kevin Gilmore, Dan Gresh, Nicholas Hunter-Jones, Karl Mayer, Brian Neyenhuis, David Hayes, and Scott Aaronson. “Demonstrating an Unconditional Separation Between Quantum and Classical Information Resources”. *Preprint*.

Scott Aaronson, Mohammad Bavarian, Toby Cubitt, Sabee Grewal, Giulio Gueltrini, Ryan O’Donnell, and Marien Raat. “Computability Theory of Closed Timelike Curves”. *Preprint*.

Sabee Grewal and Vinayak M. Kumar. “Improved Circuit Lower Bounds and Quantum-Classical Separations”. *Preprint*.

Sabee Grewal, Vishnu Iyer, William Kretschmer, and Daniel Liang. “Agnostic Tomography of Stabilizer Product States”. In: *Quantum* (2026).

Sabee Grewal, Vishnu Iyer, William Kretschmer, and Daniel Liang. “Pseudoentanglement Ain’t Cheap: Lower Bounds on the Non-Clifford Cost of Pseudoentanglement”. In: *Physical Review A* (2026). Presented at TQC 2024.

Scott Aaronson, Sabee Grewal, Vishnu Iyer, Simon Marshall, and Ronak Ramachandran. “PDQMA = DQMA = NEXP: QMA With Hidden Variables and Non-collapsing Measurements”. In: *FSTTCS 2025*.

Sabee Grewal and Justin Yirka. “The Entangled Quantum Polynomial Hierarchy Collapses”. In: *CCC 2024*.

Sabee Grewal, Vishnu Iyer, William Kretschmer, and Daniel Liang. “Efficient Learning of Quantum States Prepared With Few Non-Clifford Gates”. In: *Quantum* (2025). Presented at QIP 2024.

Sabee Grewal, Vishnu Iyer, William Kretschmer, and Daniel Liang. “Improved Stabilizer Estimation via Bell Difference Sampling”. In: *STOC 2024*. Presented at QIP 2024.

Scott Aaronson and Sabee Grewal. “Efficient Tomography of Non-Interacting Fermion States”. In: *TQC 2023*.

Sabee Grewal, Vishnu Iyer, William Kretschmer, and Daniel Liang. “Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom”. In: *ITCS 2023*. Awarded **best student paper**.

Dax Enshan Koh and Sabee Grewal. “Classical Shadows With Noise”. In: *Quantum* (2022).

RESEARCH EXPERIENCE

Student Researcher Google Quantum AI	Summer 2026
Graduate Student Visitor Simons Institute for the Theory of Computing Participated in “Summer Cluster on Quantum Computing” Host: John Wright	Summer 2025
Quantum Research Scientist Intern IBM Quantum	Summer 2024

	Graduate Student Visitor Simons Institute for the Theory of Computing Participated in “ Quantum Algorithms, Complexity, and Fault Tolerance ”	Spring 2024
	Quantum Computing Research Intern Zapata Computing	Summer 2020
	Quantum Computing Research Intern Zapata Computing	Summer 2019
OTHER EXPERIENCE	Senior Software Engineer Zapata Computing	Fall 2020
	Software Engineer II Microsoft	2017 – 2019
	Software Engineer Intern Microsoft	Summer 2016
	Software Engineer Intern Microsoft	Summer 2015
	CPU Design Intern Intel	Summer 2014
	Customer Support Agent Blizzard Entertainment	2011 – 2014
TEACHING	Lecturer , “A Short Introduction to Quantum Learning Theory” 1st AIMS Workshop and School on The Theory of Quantum Learning Algorithms	2025
	Teaching Assistant , Quantum Complexity Theory UT Austin	Spring 2025
	Teaching Assistant , Quantum Information Science I UT Austin	Fall 2023
	Head Teaching Assistant , Introduction to Computing Systems UT Austin	Fall 2021
	Teaching Assistant , Combinatorics and Graph Theory UT Austin	Spring 2021
	Head Teaching Assistant , Introduction to Computing Systems UT Austin	Fall 2019
	Teaching Assistant , Software Testing UT Austin	Spring 2019
	Teaching Assistant , Engineering Programming Languages UT Austin	Fall 2016
	Teaching Assistant , Introduction to Computing Systems UT Austin	Fall 2015
	Teaching Assistant , Software Design and Implementation UT Austin	Fall 2014

INVITED TALKS	“Demonstrating an unconditional separation between quantum and classical information resources”	
	• Mathematical Foundations of Quantum Advantage workshop	2026
	• CMU Theory Lunch	2025
	• Google Quantum AI	2025
	“Improved Circuit Lower Bounds and Quantum-Classical Separations”	
	• Columbia quantum theory lunch	2024
	“Improved Stabilizer Estimation via Bell Difference Sampling”	
	• UT Austin CS Theory Seminar	2024
SERVICE	Conference reviewer: STOC (2025–2026), FOCS (2025–2026), QIP (2023–2026), TQC (2024–2026), SODA (2025–2026), COLT (2025), CCC (2025), ITCS (2025–2026), ICALP (2024, 2026), QCTIP (2024), AQIS (2024), YQIS (2024) Journal reviewer: <i>PRX Quantum</i> (2024–2025), <i>Nature Communications</i> (2025), <i>Theory of Computing Systems</i> (2024)	